**Project Design Phase**
**Solution Architecture**

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| Date | 22 February 2026 |
| Team ID | LTVIP2026TMIDS42033 |
| Project Name | **Weather-Based Prediction of Wind Turbine Energy Output: A Next-Generation Approach to Renewable Energy Management** |
| Maximum Marks | 4 Marks |

## Solution Architecture

Solution architecture is a structured process that connects renewable energy management challenges with intelligent technological solutions. In the proposed Weather-Based Prediction of Wind Turbine Energy Output System, the architecture integrates real-time weather data, machine learning models, and an interactive dashboard to deliver accurate wind energy forecasting.

### The goals of this solution architecture are to:

● Identify and implement the most effective technology stack for real-time wind turbine energy prediction.
● Clearly describe the system structure, components, data flow, and interactions for stakeholders and evaluators.
● Define system features including weather API integration, backend processing, machine learning prediction, and analytics visualization.
● Provide technical specifications that guide development, deployment, scalability, and future enhancements of the renewable energy management platform.

### Example - Solution Architecture Diagram:

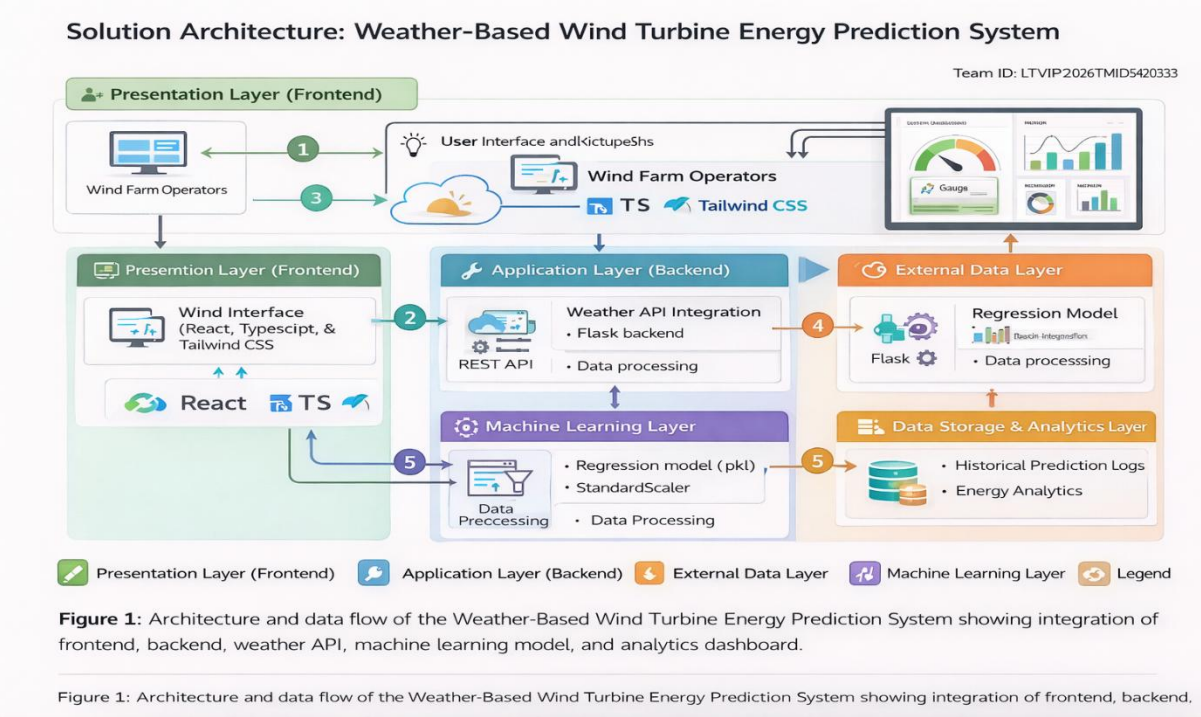


Figure 1: Architecture and data flow of the Weather-Based Wind Turbine Energy Prediction System showing integration of frontend, backend,

**Reference:** https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/